

# ReliefSphere Ai

## **1. Project Overview**

ReliefSphere AI is an AI-powered, web-based community resource redistribution platform designed to reduce resource wastage and improve equitable access to essential resources such as food, clothes, and books. It uses Artificial Intelligence (AI) and a Multi-Criteria Decision Support System to analyze donor and recipient data — resource category, urgency, geographical proximity, and organizational capacity — to recommend the most suitable recipient organization for each donation. Based on this analysis, the system provides intelligent recipient matching, delivery coordination, and priority-based emergency response. It also includes features like volunteer coordination, campaign management, notifications, and administrative oversight, creating a secure and transparent community resource redistribution ecosystem.

## **2. System Scope**

The proposed system is developed as an academic project (mini project → main project) that can later be expanded into a full-scale implementation

Current scope includes:

- Donor resource donation (food, clothes, books)
- Recipient organization verification and resource requests
- AI-based recommendation engine
- Volunteer delivery coordination
- Priority Campaign Management (emergency/community drives)
- Administrative management

Future enhancements may include:

- Explainable AI
- Mobile application (Donor, Volunteer, Recipient Org roles)
- Predictive demand analytics
- Trust scoring
- Route optimization

### **1. Target Audience**

The primary users of the system include:

- Donors
- Recipient Organizations (NGOs, orphanages, shelters, old-age homes, government schools)
- Volunteers
- System Administrators

## **2. User Roles**

- Administrator – Manages users, monitors the system, generates reports, and configures campaign/AI settings.
- Donor – Donates surplus resources, receives AI-based recipient recommendations, and tracks donation status.
- Recipient Organization – Creates resource requests, receives donations, and tracks delivery.
- Volunteer – Coordinates and updates the delivery of donated resources between donors and recipient organizations

## **3. Modules**

- 1) User Management Module – Registration, login, profile, and role management.
- 2) Donor Module – Resource donation, upload of details and images, AI recommendation view.
- 3) Recipient Organization Module – Verification, resource requests, donation acceptance/rejection.
- 4) Volunteer Coordination Module – Delivery assignment, status updates, delivery history.
- 5) AI Recommendation Module – Multi-criteria matching and ranking of recipient organizations.
- 6) Priority Campaign Management Module – Campaign creation, activation, and dynamic reprioritization.
- 7) Reports & Analytics Module – Donation statistics, campaign analytics, progress insights.
- 8) Notification Module – Status updates and delivery alerts.
- 9) Admin Dashboard Module – System monitoring, user verification, and analytics.

## **4. System Ownership**

The proposed system is developed as an **academic project** under the Department of Computer Applications.

### **System Owner:**

Academic Institution (Amal Jyothi college of engineering kanjirapally)

Developed by Amal Babu (Final Year MCA Students) Future ownership may be transferred to NGOs or community welfare organizations for practical implementation..

## **5. Questionnaire For Data Collection**

### ***Target Users: Donors***

#### **1. How often do you have surplus resources available to donate?**

Occasionally, once every 1-2 months

#### **2. What is your biggest challenge when trying to donate?**

Not knowing which organization actually needs what I have.

**3. Do you currently know which NGO/shelter/school needs what you're donating?**

No, usually not

**4. Would AI-based recommendations help?**

Yes, that would save a lot of time.

**5. Which feature would you find most useful?**

Knowing the urgency level of each organization's need.

**6. How do you currently find organizations to donate to?**

Through word of mouth or social media posts

**7. Would a single platform for donating and tracking help?**

Yes, definitely

**8. Would you donate more often if the process were simpler?**

Yes.

***Target Users: Recipient Organizations***

**1. How often does your organization face a resource shortage?**

Frequently, especially during certain seasons.

**2. Biggest concern about receiving timely donations?**

Unpredictability — donations don't always match what we actually need.

**3. Do you maintain records of your requirements?**

Yes, but manually / informally.

**4. Would need-matched donations help?**

Yes, greatly

**5. What is the Most useful feature?**

Priority alerts during emergencies.

**6. How do you currently source donations?**

Through personal contacts, local donors, and occasional NGO networks

**7. Would a priority/emergency campaign system help?**

Yes, especially during disasters or seasonal shortages.

**8. Would verified donor info increase trust?**

Manual registers or spreadsheets.

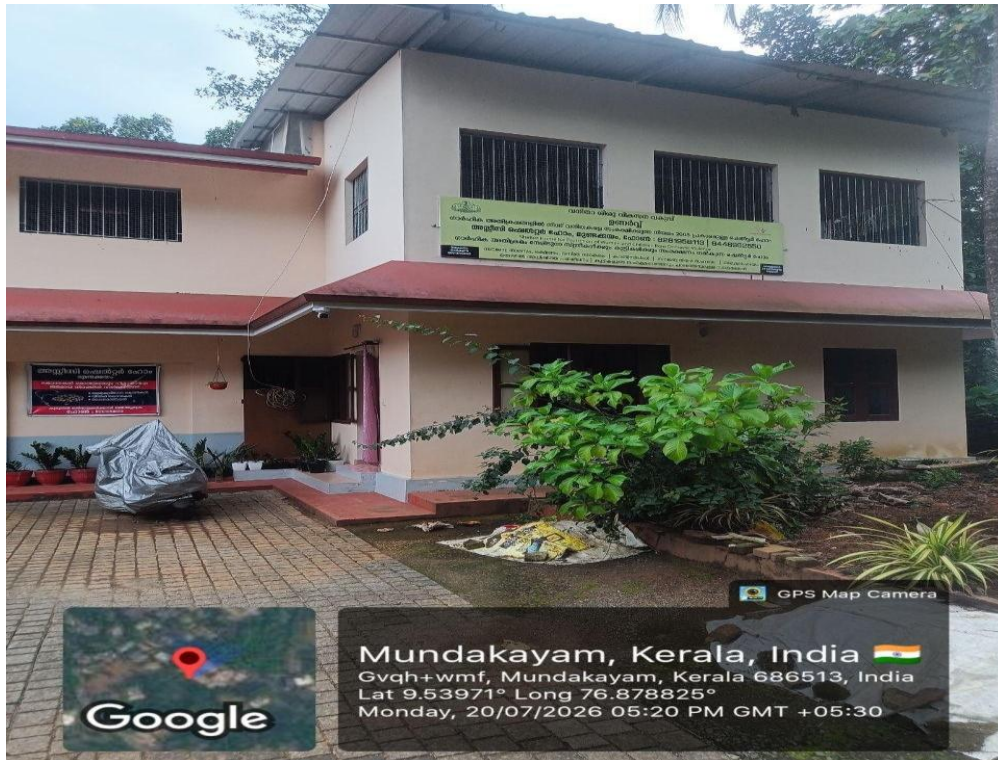
**9. Would you recommend such a platform?**

Yes, Definitely

## 6. Data Collection Contact

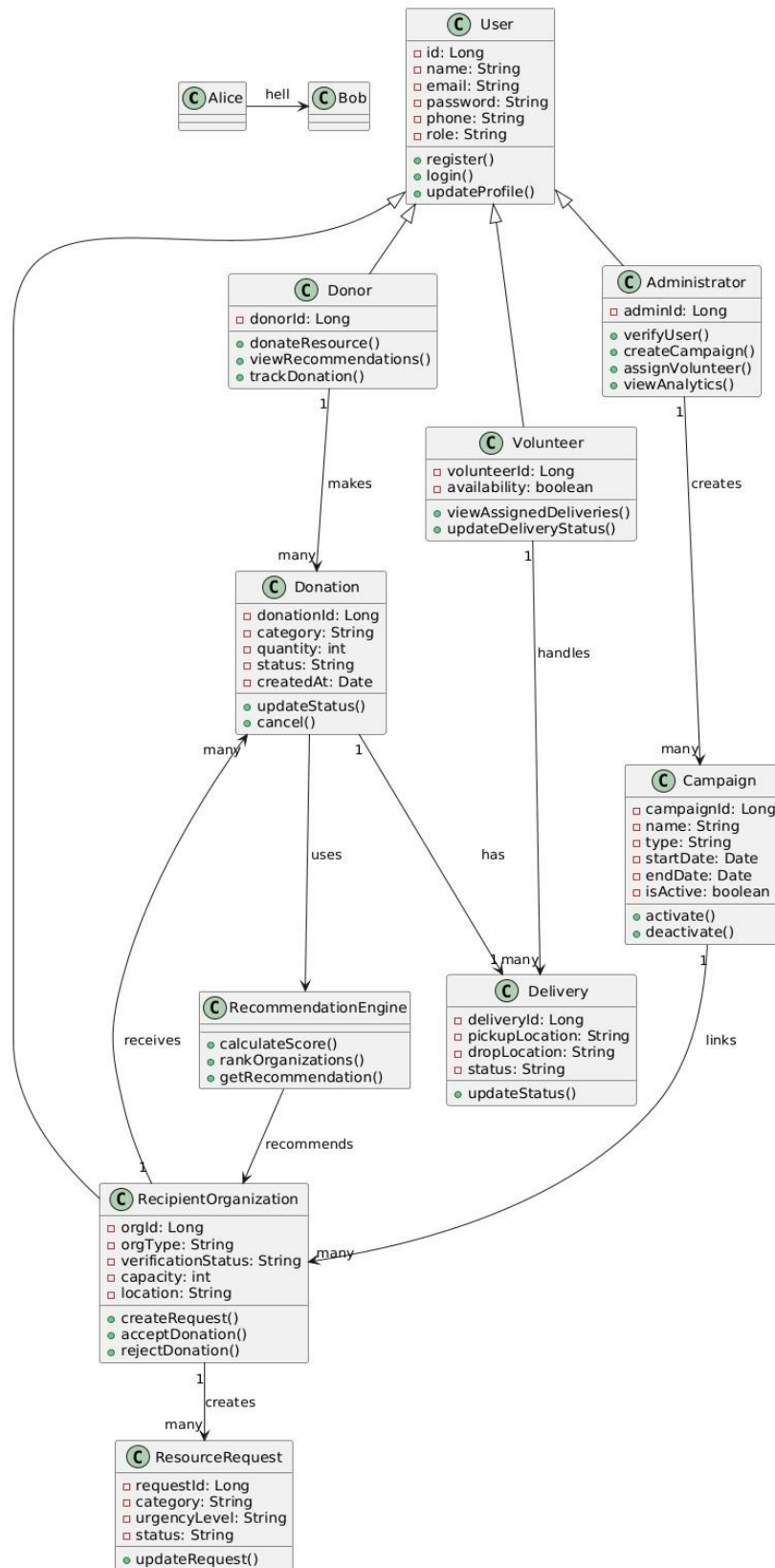
Name : Sr.Mary (Recipient Organization representative)

Phone : +91 62386 77516

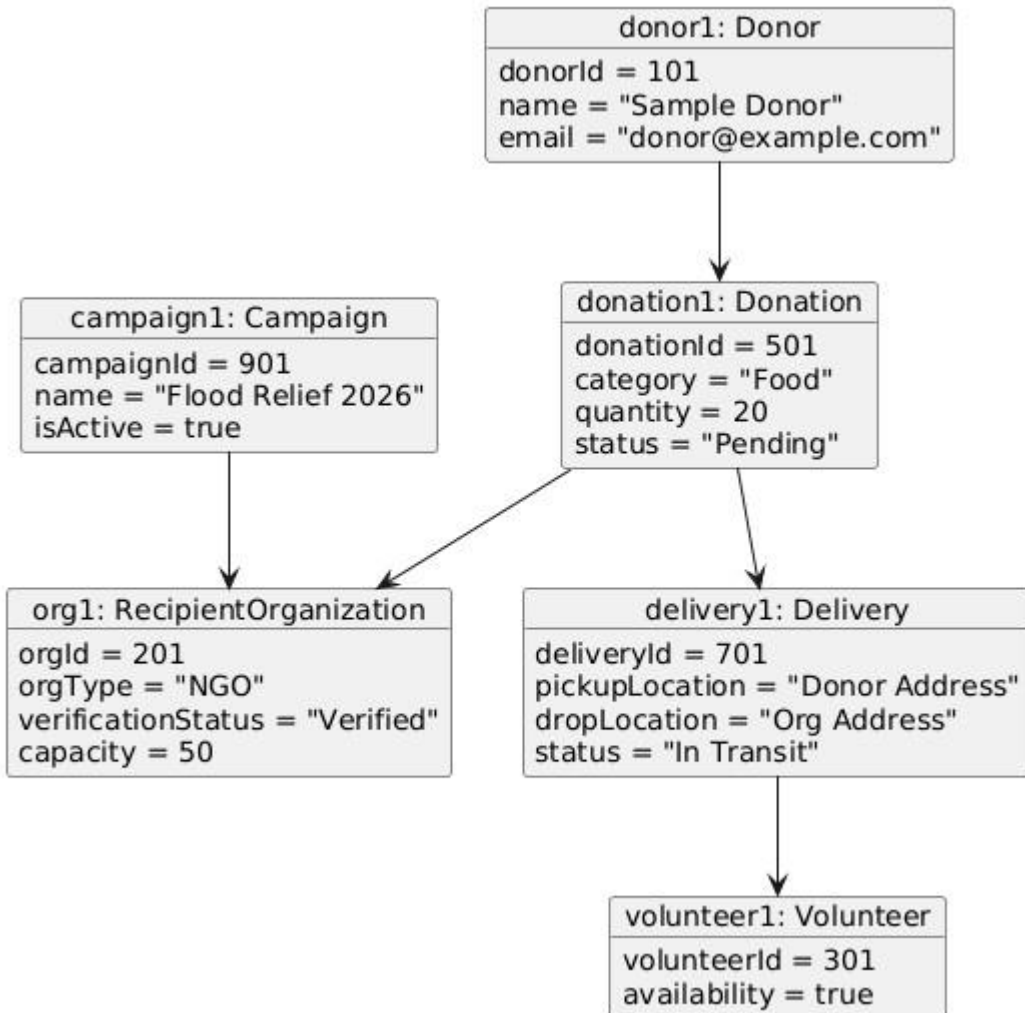


## UML DIAGRAMS

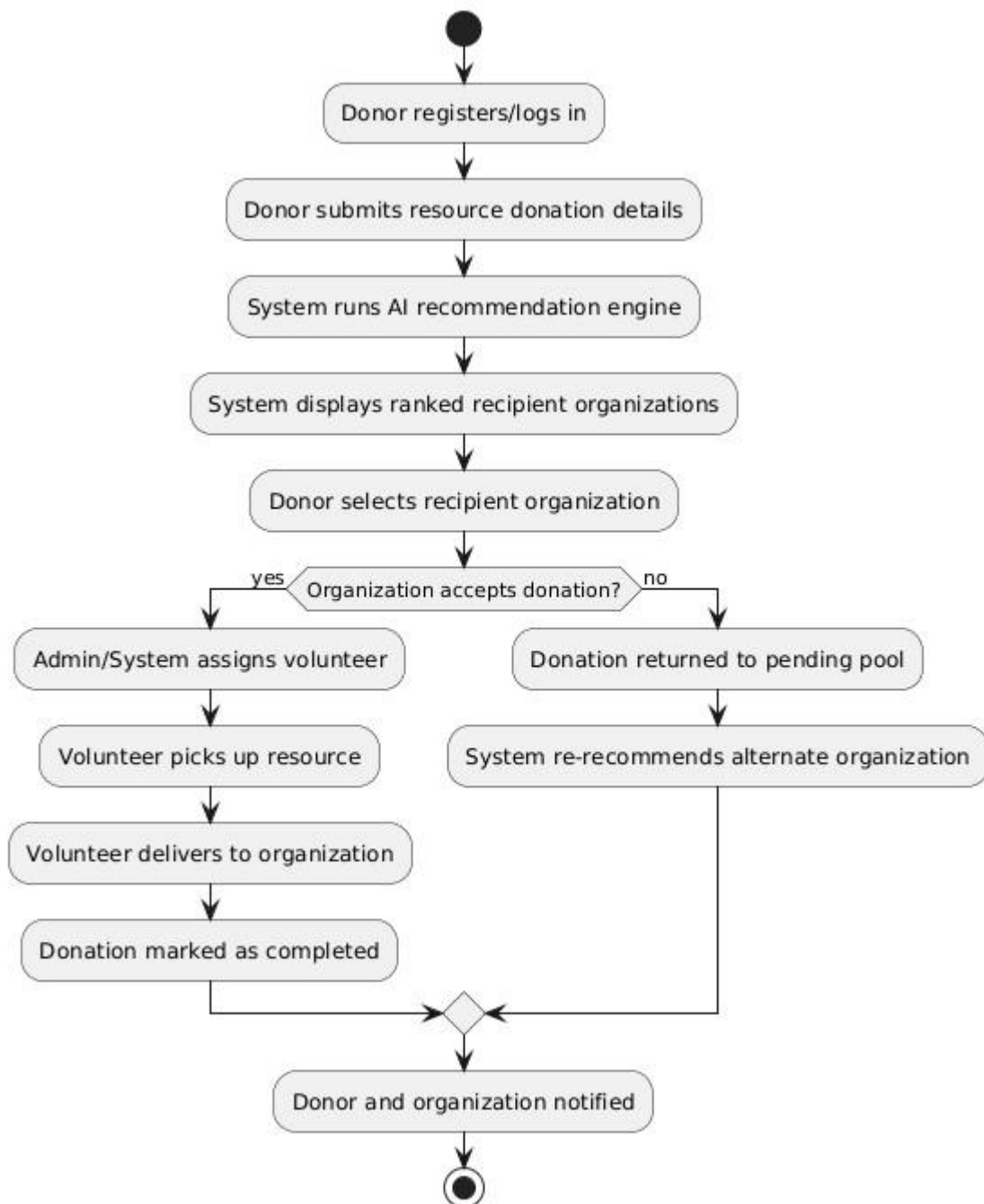
### class Diagram



## Deployment Diagram

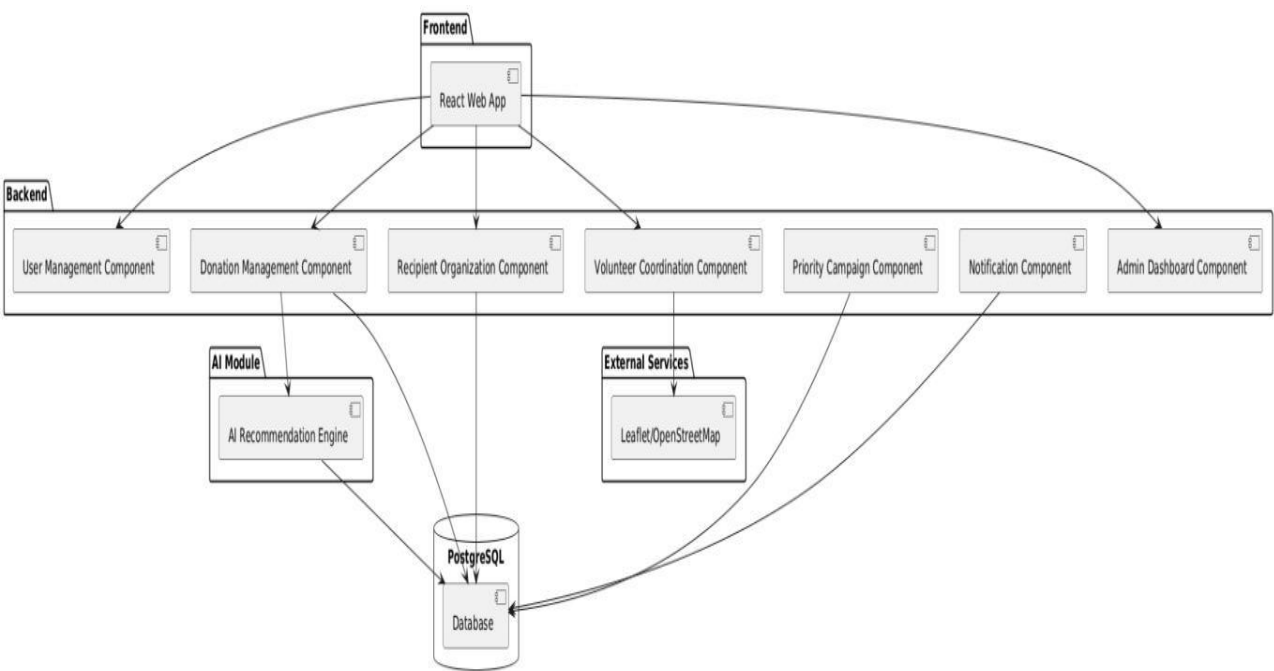


## Activity Diagram

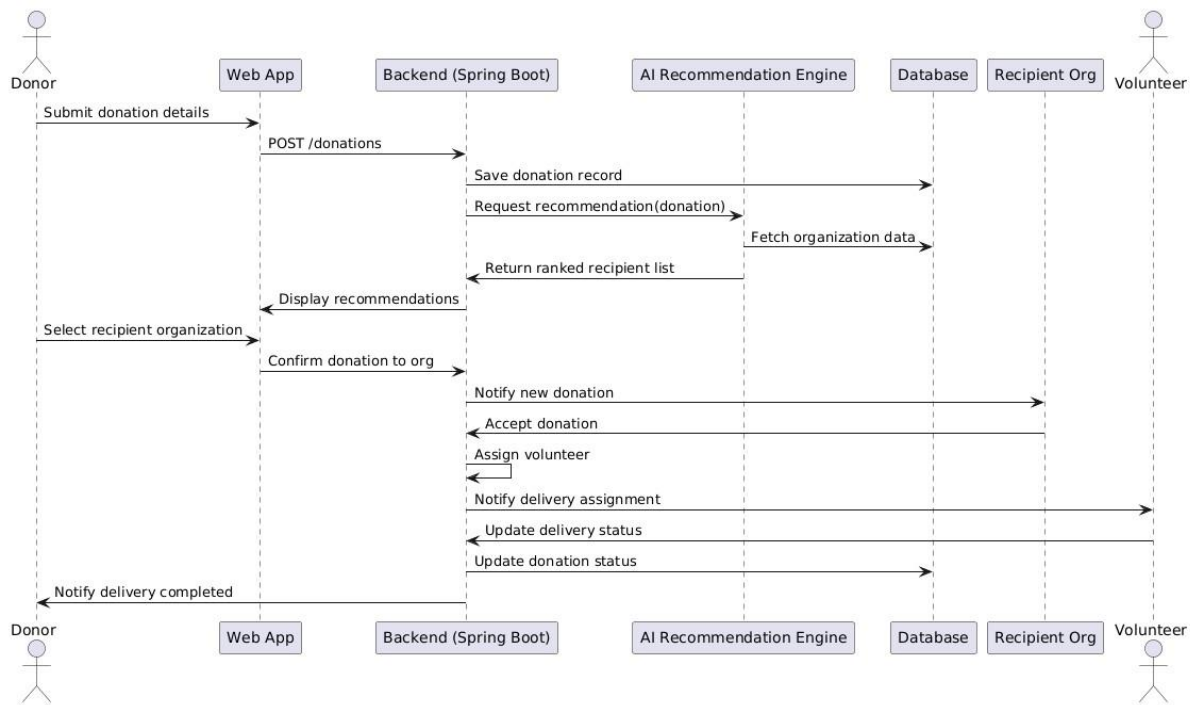




# Component Diagram

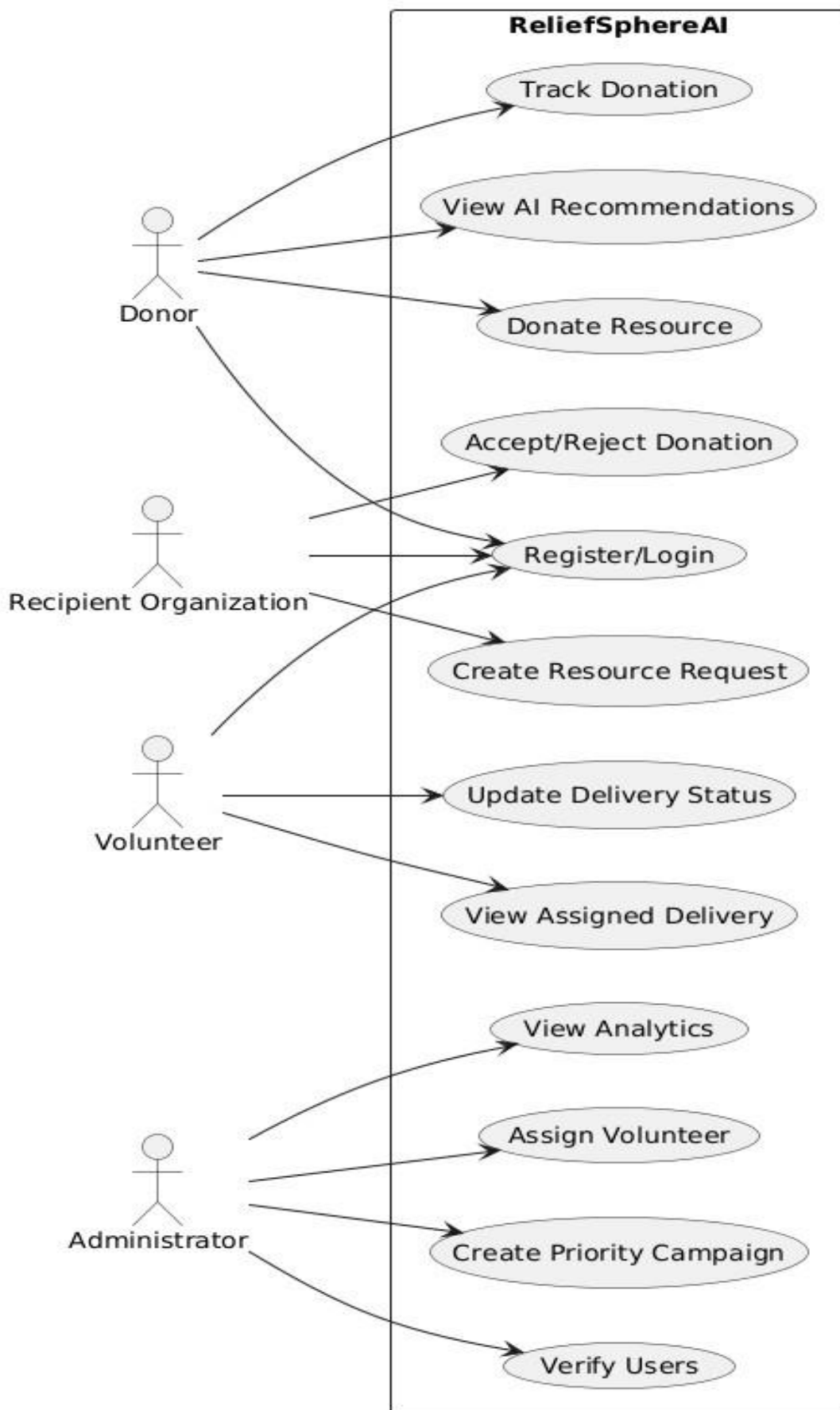


# Sequence Diagram

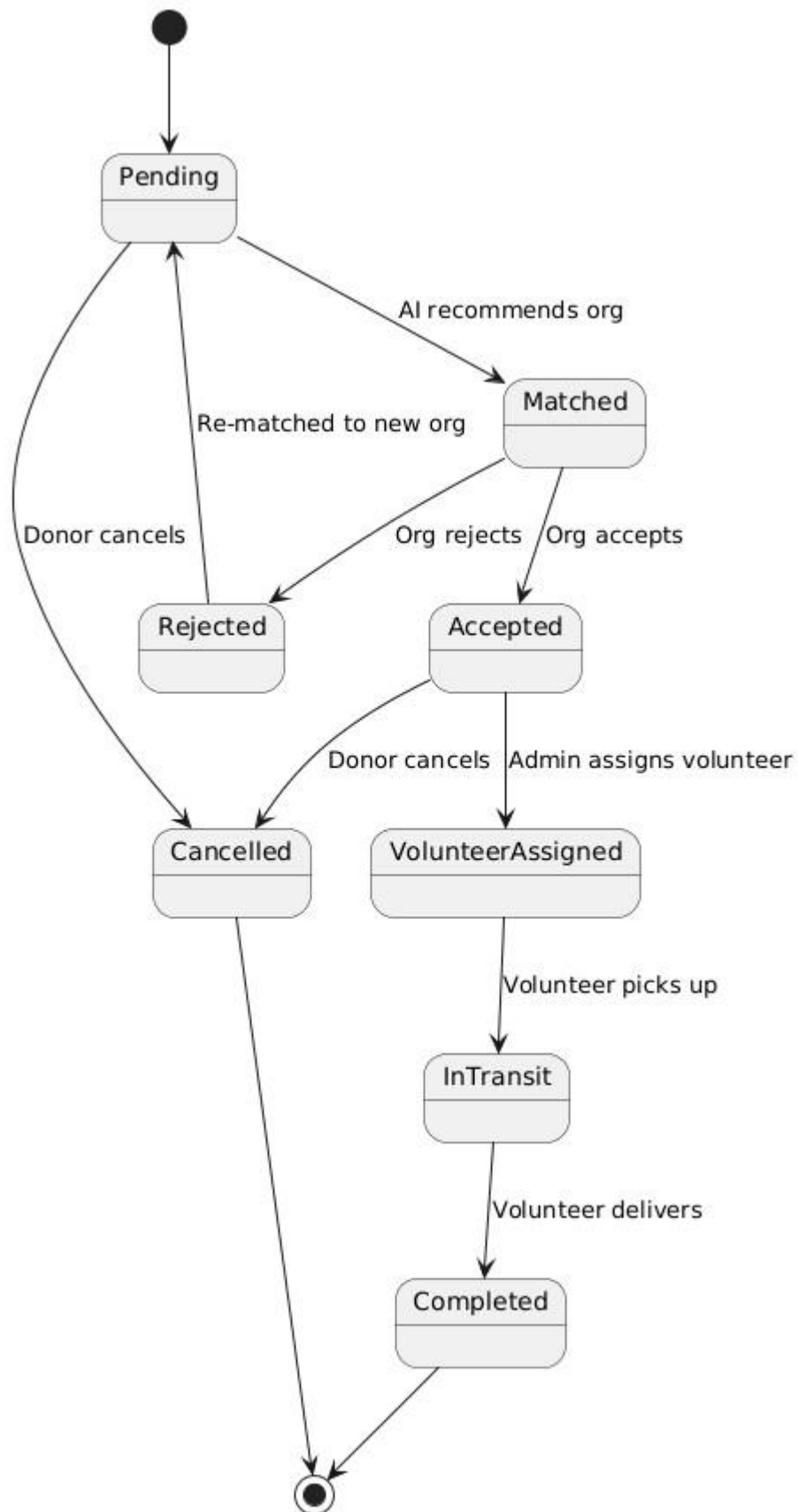




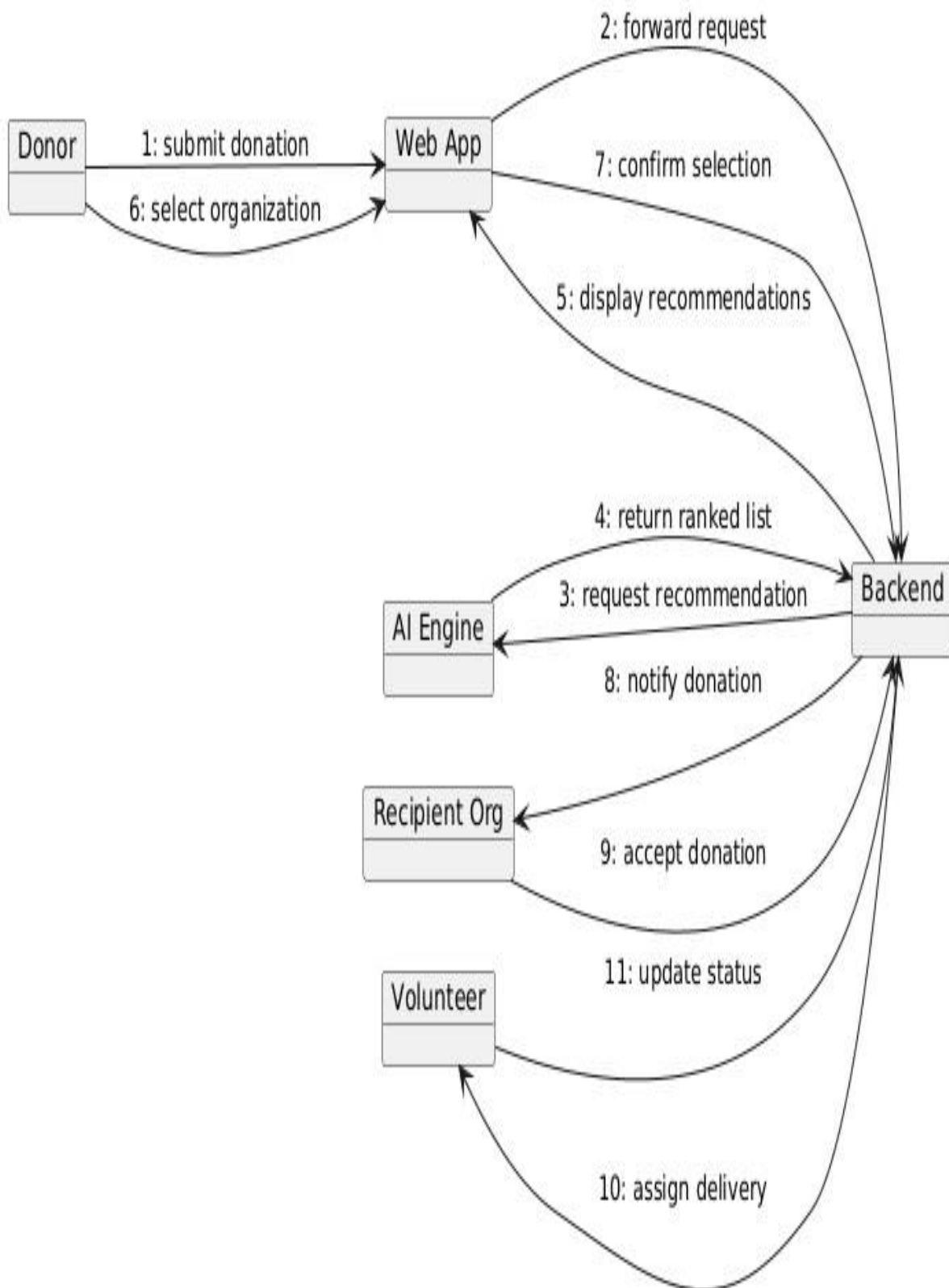
## Usecase Diagram



## StateChart Diagram



## Collaboration Diagram



# Object Diagram

